# sandbox-api

API Reference Documentation

Version 1.0.0 • March 2026 • Zoho CRM + RDP Storage

| 15 Endpoints | REST Architecture | OAuth2 Zoho Auth | SOLID Principles |
|---|---|---|---|

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## 1. Overview & Base URL

sandbox-api is a production-grade Node.js REST API built with Express. It fetches data from Zoho CRM using OAuth2, stores records as JSON on the RDP server disk, downloads attachments, and provides paginated + chunked access for large datasets. Built with SOLID principles, ESM modules, and daily-rotating Winston logs.

```
// Base URL
http://<rdp-server>:3000

// Versioned API base
http://<rdp-server>:3000/api/v1

// Swagger UI (interactive docs)
http://<rdp-server>:3000/api-docs
```

## 2. Authentication

All `/api/v1/*` endpoints require an API key passed in the request header. The `/health` and `/api-docs` routes are public and require no authentication.

```
// Required header on all /api/v1/* requests
X-API-Key: your_api_key_here

// Unauthorized response (401)
{
"success": false,
"error": {
"code": "UNAUTHORIZED",
"message": "Invalid or missing API key."
}
}
```

Set `API_KEY` in your `.env` file. Generate a secure key with: `node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"`

## 3. Error Handling

All errors follow a consistent JSON structure. Every response includes a `requestId` for tracing in logs.

```
// Error response structure
{
"success": false,
"requestId": "a1b2c3d4-e5f6-...",
"error": {
"code": "VALIDATION_ERROR",
"message": "module is required"
}
}
```

| HTTP | Code | Meaning |
|---|---|---|
| 400 | VALIDATION_ERROR | Invalid or missing request parameters |
| 401 | UNAUTHORIZED | Missing or invalid X-API-Key header |
| 404 | NOT_FOUND | Resource or route not found |
| 409 | CONFLICT | Sync already in progress for this module |
| 429 | RATE_LIMIT_EXCEEDED | Too many requests — slow down |
| 500 | INTERNAL_ERROR | Unexpected server error — check logs |

## 4. Rate Limiting

Two rate limit tiers protect the API from overuse. Limits are per IP address and reset every 60 seconds. Exceeding the limit returns HTTP 429.

| Tier | Limit | Applies To |
|---|---|---|
| Standard | **100 req / min** | All `/api/v1/*` endpoints |
| Strict | **10 req / min** | `POST /sync`, `POST /attachments/.../download` |

# 5. Health Check

| | | |
|---|---|---|
| GET | /health | Public liveness probe — no auth required |

```
// 200 OK
{
"status": "ok",
"version": "v1",
"env": "development",
"timestamp": "2026-03-10T12:00:00.000Z"
}
```

# 6. CRM Records

## 6.1 Fetch Records (Live)

| | | |
|---|---|---|
| GET | /api/v1/crm/records | Fetch records live from Zoho CRM. Results cached 5 min. |

**Query Parameters**

| Parameter | In | Type | Required | Description |
|---|---|---|---|---|
| module | query | string | Yes | CRM module: Leads, Contacts, Accounts, Deals, Tasks |
| page | query | integer | No | Page number. Default: 1 |
| per_page | query | integer | No | Records per page. Default: 50. Max: 200 |
| fields | query | string | No | Comma-separated field names to return |
| sort_by | query | string | No | Field name to sort by |
| sort_order | query | string | No | Sort direction: asc or desc. Default: desc |

**Responses**

| Status | Description |
|---|---|
| 200 | Records fetched successfully. Returns data array + Zoho pagination info. |
| 400 | Validation error — invalid module or parameter value. |
| 401 | Missing or invalid X-API-Key. |

## 6.2 Fetch Single Record

| | | |
|---|---|---|
| GET | /api/v1/crm/records/{module}/{id} | Fetch one CRM record by its Zoho ID. |

### Path Parameters

| Parameter | In | Type | Required | Description |
|---|---|---|---|---|
| `module` | path | `string` | Yes | CRM module name e.g. Leads |
| `id` | path | `string` | Yes | Zoho record ID e.g. 1113843000000696206 |

### Responses

| Status | Description |
|---|---|
| 200 | Record found and returned. |
| 404 | Record not found in the specified module. |

# 7. Sync Endpoints

Sync endpoints fetch all records for a module from Zoho CRM and persist them to the RDP disk in two formats: a single `latest.json` for quick access, and chunked files (500 records/file) for memory-safe access to large datasets. A sync queue prevents concurrent syncs on the same module.

## 7.1 Sync Status

**GET** `/api/v1/crm/sync/status` — Check which modules are currently syncing.

```
// 200 OK — no active syncs
{ "success": true, "activeSyncs": 0, "modules": [] }

// 200 OK — Leads sync in progress
{ "success": true, "activeSyncs": 1, "modules": ["Leads"] }
```

## 7.2 Sync Module

**POST** `/api/v1/crm/sync` — Fetch all records for a module and save to RDP disk.

**Request Body — application/json**

| Parameter | In | Type | Required | Description |
|---|---|---|---|---|
| `module` | body | `string` | Yes | Module to sync: Leads, Contacts, Accounts, Deals |
| `overwrite` | body | `boolean` | No | Force re-sync even if data is fresh (< 30 min old). Default: false |

**Responses**

| Status | Description |
|---|---|
| 200 | Sync complete. Returns count, chunkCount, latestPath, syncedAt. |
| 200 (fresh) | Data is still fresh (< 30 min). Use overwrite:true to force. |
| 400 | Validation error — invalid module. |
| 409 | Sync already in progress for this module. Try again shortly. |
| 429 | Strict rate limit exceeded — max 10 requests/min. |

```
// Request body
{ "module": "Leads", "overwrite": true }

// 200 OK — sync complete
{
```

```
"success": true,
"message": "Sync complete for Leads",
"module": "Leads",
"count": 9,
"chunkCount": 1,
"latestPath": "./storage/crm-data/Leads/latest.json",
"syncedAt": "2026-03-10T12:00:00.000Z"
}

// 409 — already syncing
{
"success": false,
"message": "Sync for Leads is already in progress.",
"startedAt": "2026-03-10T11:59:50.000Z"
}
```

# 8. Storage Endpoints

Storage endpoints read data that was previously saved by `POST /sync`. Two read modes are available: full JSON (small datasets) and paginated chunked reads (memory-safe for large datasets).

| Endpoint | Use When |
|---|---|
| `GET /storage/{module}` | Small datasets — loads entire latest.json into memory |
| `GET /storage/{module}/paged` | Large datasets — reads one chunk at a time, never loads all records |
| `GET /storage/{module}/meta` | Just need the count / chunk info without loading any records |

## 8.1 Get Latest Stored Data

**GET** `/api/v1/crm/storage/{module}` — Return full latest.json for a module.

| Status | Description |
|---|---|
| 200 | Returns { module, fetchedAt, count, records[] } |
| 404 | No data stored yet — run POST /sync first. |

## 8.2 Get Paged Data (Memory Safe)

**GET** `/api/v1/crm/storage/{module}/paged` — Paginated read — loads one chunk at a time.

| Parameter | In | Type | Required | Description |
|---|---|---|---|---|
| `module` | path | `string` | Yes | CRM module name e.g. Leads |
| `page` | query | `integer` | No | Page number. Default: 1 |
| `pageSize` | query | `integer` | No | Records per page. Default: 50. Max: 500 |

```
// 200 OK
{
"success": true,
"module": "Leads",
"records": [...],
"page": 1,
"pageSize": 50,
"totalCount": 5000,
"totalPages": 100,
```

```
"hasMore": true,
"fetchedAt": "2026-03-10T12:00:00.000Z"
}
```


### 8.3 Get Storage Meta

**GET** `/api/v1/crm/storage/{module}/meta` — Return metadata only — no records loaded.


```
{
"success": true,
"module": "Leads",
"fetchedAt": "2026-03-10T12:00:00.000Z",
"totalCount": 5000,
"chunkCount": 10,
"chunkSize": 500
}
```

## 8.4 List Snapshots

| | | |
|---|---|---|
| GET | /api/v1/crm/storage/{module}/snapshots | List all saved snapshot filenames for a module. |


```
{
"success": true,
"module": "Leads",
"count": 3,
"snapshots": [
"2026-03-10T10-09-55.json",
"2026-03-09T08-00-00.json",
"2026-03-08T14-30-00.json"
]
}
```


## 8.5 Get Snapshot Data

| | | |
|---|---|---|
| GET | /api/v1/crm/storage/{module}/snapshots/{filename} | Return full data from a specific snapshot file. |

| Parameter | In | Type | Required | Description |
|---|---|---|---|---|
| module | path | string | Yes | CRM module name e.g. Leads |
| filename | path | string | Yes | Snapshot filename e.g. 2026-03-10T10-09-55.json |

| Status | Description |
|---|---|
| 200 | Returns full snapshot data: { module, fetchedAt, count, records[] } |
| 404 | Snapshot file not found. |

# 9. Attachment Endpoints

Attachment endpoints interact with files attached to Zoho CRM records. Attachments are not included in the standard record response — they require separate API calls. Downloaded files are saved to `storage/attachments/{module}/{recordId}/` on the RDP disk.

## 9.1 List Attachments

**GET** `/api/v1/crm/attachments/{module}/{recordId}` — List all attachments for a CRM record.

| Parameter | In | Type | Required | Description |
|---|---|---|---|---|
| `module` | path | `string` | Yes | CRM module name e.g. Leads |
| `recordId` | path | `string` | Yes | Zoho record ID e.g. 1113843000000696206 |

```
// 200 OK — 2 attachments found
{
"success": true,
"module": "Leads",
"recordId": "1113843000000696206",
"count": 2,
"attachments": [
{ "id": "1113843000000825007", "File_Name": "document.pdf", "Size": "24926", ... },
{ "id": "1113843000000825001", "File_Name": "contract.pdf", "Size": "32463", ... }
]
}

// 200 OK — no attachments
{ "success": true, "module": "Leads", "recordId": "...", "count": 0, "attachments": [] }
```

## 9.2 Download All Attachments

**POST** `/api/v1/crm/attachments/{module}/{recordId}/download` — Download all attachments to RDP disk.

No request body required. Files are streamed directly to disk — they are never fully loaded into memory.

| Status | Description |
|---|---|
| 200 | All attachments downloaded. Returns file paths and sizes. |
| 200 (empty) | No attachments found for this record. Returns total: 0. |
| 429 | Strict rate limit exceeded — max 10 requests/min. |

```
// 200 OK — files downloaded to RDP disk
{
"success": true,
"recordId": "1113843000000696206",
"module": "Leads",
"total": 2,
"downloaded": 2,
"attachments": [
{
"attachmentId": "1113843000000825007",
"fileName": "document.pdf",
"filePath": "F:\\storage\\attachments\\Leads\\1113843000000696206\\document.pdf",
"size": 24926
}
]
}
```

# 10. Environment Variables

Copy `.env.example` to `.env` and fill in all required values before starting the server.

### Server

| Variable | Default | Required | Description |
|---|---|---|---|
| `NODE_ENV` | `development` | No | Environment: development or production |
| `PORT` | `3000` | No | HTTP port the server listens on |
| `API_VERSION` | `v1` | No | API version prefix used in all routes |

### Security

| Variable | Default | Required | Description |
|---|---|---|---|
| `API_KEY` | — | Yes | Secret key sent in X-API-Key header |
| `RATE_LIMIT_WINDOW_MS` | `60000` | No | Rate limit window in milliseconds |
| `RATE_LIMIT_MAX_REQUESTS` | `100` | No | Max requests per window (standard tier) |

### Zoho CRM

| Variable | Default | Required | Description |
|---|---|---|---|
| `ZOHO_CLIENT_ID` | — | Yes | Client ID from Zoho API Console |
| `ZOHO_CLIENT_SECRET` | — | Yes | Client Secret from Zoho API Console |
| `ZOHO_REFRESH_TOKEN` | — | Yes | Refresh token (permanent) from OAuth2 flow |
| `ZOHO_REDIRECT_URI` | `https://www.zohoapis.in/crm/v2` | No | Must match Zoho console setting |
| `ZOHO_ACCOUNTS_URL` | `https://accounts.zoho.in/oauth/v2/token` | No | Zoho IN region token URL |

### Storage

| Variable | Default | Required | Description |
|---|---|---|---|
| `STORAGE_BASE_PATH` | `./storage/crm-data` | No | Root path for CRM JSON storage |
| `ATTACHMENT_STORAGE_PATH` | `./storage/attachments` | No | Root path for downloaded attachments |

## Logging

| Variable | Default | Required | Description |
|---|---|---|---|
| `LOG_LEVEL` | `info` | No | Winston log level: error, warn, info, debug |
| `LOG_DIR` | `./logs` | No | Directory for daily-rotating log files |